

BANARAS HINDU UNIVERSITY
B.A./B.Sc. (Hons.) Part-III Examination, 2014-15

Paper No. Paper-VII: Industrial Statistics and Operations Research (Old Course)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Question No. 1 is compulsory. Answer **four** more questions selecting one from each Unit. Symbols have their usual meanings.*

Question 1 (Compulsory). Answer in brief of the following:

- (a) Define a saddle point in game theory. [2]
- (b) Define Operations Research (O.R.). [2]
- (c) In which way does a transportation problem differ from an assignment problem? [2]
- (d) Define Reliability in reliability engineering. [2]
- (e) Who is known as the father of game theory? [2]
- (f) Who developed the transportation problem? [2]
- (g) Define a parallel system with a suitable example. [2]
- (h) For which probability distribution is the hazard rate constant? [2]
- (i) Define non-degenerate feasible solution in linear programming. [2]
- (j) Write the dual of the following Linear Programming Problem (LPP):

$$\begin{aligned} &\text{Minimize } Z = 15x_1 + 10x_2 \\ &\text{subject to: } 5x_1 + 3x_2 \geq 5 \\ &\quad \quad \quad x_1 + 5x_2 \geq 12 \\ &\quad \quad \quad x_1, x_2 \geq 0 \end{aligned}$$

[2] 2. Explain how and why Operations Research (O.R.) methods are valuable in aiding executive decision making. Discuss the phases of an O.R. study. [12.5]

3. Solve the following Linear Programming Problem (LPP) using the Simplex method:

$$\begin{aligned} &\text{Minimize } Z = x_1 - 3x_2 + 2x_3 \\ &\text{subject to: } 3x_1 - x_2 + 2x_3 \leq 7 \\ &\quad \quad \quad -2x_1 + 4x_2 \leq 12 \\ &\quad \quad \quad -4x_1 + 3x_2 + 8x_3 \leq 10 \\ &\quad \quad \quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

[12.5]

UNIT - II

4. Describe different methods for finding an initial basic feasible solution of a Transportation Problem. Solve the following Transportation Problem to find the minimum transportation cost:

	I	II	III	IV	Availability
A	5	1	3	3	34
B	3	3	5	4	15
C	6	4	4	3	12
D	4	1	4	2	19
Requirement	21	25	17	17	80

[12.5]

5. What are inventory models? Describe the Economic Lot Size (ELS) model with different rates of demand in different production cycles, infinite production rate, and having no shortages.

[12.5]

UNIT - III

6.

(k) What do you mean by the following terms in game theory?

- (i) Competitive game
- (ii) Pay-off matrix
- (iii) Pure and mixed strategies

[6.5]

(l) Solve the following game:

		Player B	
		I	II
Player A	I	2	5
	II	4	1

[6] 7. What is a rectangular game? How will you find its graphical solution? Explain the maximin and minimax criterion as applied to the theory of games.

[12.5]

UNIT - IV

8.

(m) Discuss the relationship among $f(t)$ (probability density function), $F(t)$ (cumulative distribution function), $R(t)$ (reliability function), and $h(t)$ (hazard rate function). [6.5]

(n) Discuss the nature of the hazard rate function $h(t)$ in Weibull distribution and Gamma distribution. [6]

[6.5]

(o) Define a parallel system and a series system in reliability theory, and derive their reliability expressions. Give a typical example of each. [6.5]

[6.5]

- (p) Describe the exponential distribution as a failure model. Prove that the hazard rate function $h(t)$ is constant for the exponential distribution. [6]